

FOR_04 — Digital Evidence, the Forensic Process, Mobile Computing & Mobile Forensics

Covers TP-II Unit 8 (Digital Evidences) and Unit 7 (Mobile Computing)

TP-II Unit 8 is the "**principles and process**" capstone of the elective — it takes everything from FOR_03 (imaging, hashing, recovery, tracing) and frames it as a disciplined, court-defensible **process** with named principles (ACPO), a named **model** (identify→...→present), and a deliverable (the **report** and **expert testimony**). Unit 7 is two halves welded together: the **mobile computing** engineering (cells, protocols, mobile databases, M-business) *and* **mobile forensics** (SIM/IMEI, CDR, acquisition levels, Android/iOS). The engineering half overlaps CORE networks; the forensic half is unique to this elective and high-yield.

Cross-references: imaging/hashing/write blockers/carving/signatures/steganalysis are treated in depth in **FOR_03 §5–§6** and file systems/registry/slack in **FOR_02**. Here they are **summarised and framed as process/evidence**, not repeated in full — the exam wants you to *place* each technique inside the process and the ACPO principles.

△ **Indian-law accuracy:** §65B IEA 1872 → §63 BSA 2023 (cite both); IT Act sections per FOR_03; flag any uncertain section number △ *verify*. Never invent a section number or a case name.

PART A — DIGITAL EVIDENCE AND THE FORENSIC PROCESS (TP-II Unit 8)

§1. Digital Evidence — Concept and Characteristics

1.1 Concept

Digital evidence = any information of probative value that is stored or transmitted in binary form and may be relied upon in court. It is a **silent, invisible witness**: unlike a bloodstain you can hold, digital evidence must be *made visible* by tools, and it is uniquely **fragile, easily altered, easily copied, and time-sensitive**.

Characteristics (learn as a list — a common 5-mark question):

1. **Latent** — not visible to the naked eye; needs tools/software to be perceived.
2. **Fragile / volatile** — easily changed, damaged or destroyed (a single mount writes to it).
3. **Easily duplicated** — a perfect **bit-for-bit copy** is possible (and is the working method).
4. **Alterable and time-sensitive** — timestamps, RAM, logs change or vanish (order of volatility).
5. **Can transcend jurisdictions** — spread across devices, servers, countries.
6. **Voluminous** — terabytes; needs triage and selective analysis.
7. **Machine/format dependent** — needs the right tool to interpret.

Admissibility requirements (the four the examiner wants): digital evidence must be **Authentic** (it is what it claims to be — proven by hash + chain of custody), **Reliable/accurate** (sound tools and methods), **Complete** (not cherry-picked), and **legally obtained & admissible** (lawful seizure + **§65B IEA / §63 BSA certificate**). Some books add **"believable"** (understandable to the court). Cite the certificate requirement explicitly.

1.2 Digital-evidence storage devices (where it lives)

HDD, SSD, USB flash, SD/microSD, eMMC/UFS (phones), optical (CD/DVD/Blu-ray), magnetic tape (LTO), RAM/volatile memory, firmware/NVRAM, RAID/NAS/SAN, and **cloud**. (Geometry, SSD/TRIM problem and

RAID levels are in **FOR_02 §2**; cloud in **FOR_03 §10**.)

1.3 Order of volatility (recap — see **FOR_03 §4.3** for the full list + **RFC 3227**)

Most → least volatile: **registers/cache** → **routing/ARP/process tables & network connections** → **RAM** → **temp/swap files** → **disk** → **remote logs** → **physical topology** → **archival backups**. Collect most-volatile first. This is the operational form of the **Law of Progressive Change** (**FOR_01 §1.5**).

§2. Basic Rules / Principles for Cyber Forensics — the ACPO Principles

This is a near-certain question. Learn the **four ACPO principles** verbatim in substance. "ACPO" = the (UK) **Association of Chief Police Officers**; its *Good Practice Guide for Digital Evidence* is the most-cited statement of digital-forensic first principles worldwide.

#	Principle (substance)	What it means in practice
1	No action taken should change data held on a computer/ storage medium that may be relied upon in court	Use write blockers ; image, don't work on the original; don't boot the evidence machine
2	If a person must access the original data, they must be competent to do so and able to explain the relevance and implications of their actions	Live acquisition (RAM, unlocked encryption) is allowed when justified by a competent examiner — document <i>why</i>
3	An audit trail (record of all processes applied) should be created and preserved; an independent third party should be able to repeat them and reach the same result	Contemporaneous notes, tool versions, hashes, chain of custody → reproducibility
4	The person in charge of the investigation has overall responsibility for ensuring these principles are followed	Accountability sits with the officer/lead

How to deploy in the exam: whenever a question asks "how do you ensure evidence is admissible / defensible", answer through the four ACPO principles + hashing + chain of custody + §65B/§63. It is a reusable spine for many 10/15-mark answers.

Related frameworks to name-drop: **RFC 3227** (evidence collection order), **NIST SP 800-86** (guide to integrating forensics into incident response), **ISO/IEC 27037** (identification, collection, acquisition, preservation of digital evidence) △ verify these document numbers before quoting them as exact.

§3. The Digital Forensic Process Model — the 6 phases (draw it)

Guaranteed descriptive question: "Explain the phases of the digital forensic process / cyber forensic investigation." Learn the six phases, one line each, plus what could go wrong at each.

IDENTIFICATION → PRESERVATION → COLLECTION → EXAMINATION → ANALYSIS → PRESENTATION
(find it) (protect it) (acquire it) (extract it) (interpret) (report/testify)
_____ documentation & chain of custody run through ALL phases

Phase	Goal	Key activities	Failure mode
1. Identification	Recognise what and where the evidence is	Identify devices, data sources, scope, legal authority; assess power state; identify volatile vs non-volatile	Missing a device (NAS, phone, cloud); scope creep beyond warrant
2. Preservation	Prevent any change	Isolate/secure scene, write blockers, order of volatility, imaging with hashing , chain of custody, Faraday for phones	Bootting the machine; no write blocker; remote wipe
3. Collection / Acquisition	Obtain the data	Bit-stream/logical/sparse imaging; live vs static; hash source & image	Wrong acquisition type; unverified image
4. Examination	Extract the relevant data from the mass	Recover deleted files, carve, decrypt, keyword search, filter with hash sets, parse artifacts	Overlooking slack/unallocated/ADS; missing hidden data
5. Analysis	Interpret and correlate → conclusions	Timeline reconstruction, link analysis, attribution, corroboration across sources	Over-claiming; confirmation bias (FOR_01 §3.4)
6. Presentation / Reporting	Communicate to the court	Clear report, exhibits, expert testimony , §65B/§63 certificate	Jargon; unsupported opinion; broken chain exposed in cross

Some models use different labels (**Kruse & Heiser: Acquire → Authenticate → Analyse; DFRWS: Identification → Preservation → Collection → Examination → Analysis → Presentation → Decision; NIST SP 800-86: Collection → Examination → Analysis → Reporting**). If asked for a model, give the **six-phase** version and mention that NIST condenses it to **four** (Collection, Examination, Analysis, Reporting). Naming one alternative model earns a mark.

§4. Data Acquisition Methods

4.1 By completeness

Method	What it captures	Use / trade-off
Physical (bit-stream) acquisition	Every sector — allocated + unallocated + slack + HPA/DCO	The forensic default; enables deleted-file recovery & carving; largest, slowest
Logical acquisition	Only the active files the file system reports	Faster/smaller; no deleted data ; used for huge systems or targeted, authorised subsets
Sparse acquisition	A logical copy of selected data + some deleted/unallocated fragments relevant to the case, without the whole device	Middle ground for very large data sets; capture only what is relevant (common in mobile/enterprise)
Targeted / triage	Specific artifacts (a few files, memory) captured quickly on scene	Live triage; used to decide priorities

4.2 By system state

Live (dynamic) acquisition	Static (dead-box) acquisition
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System	Running	Powered off
Gets	RAM, running processes, network state, unlocked encrypted volumes , cloud sessions	Full disk image, repeatable
Risk	Alters some state (justify per ACPO P2); rootkit may lie	Loses all volatile evidence

(Full treatment in **FOR_03 §4.4**; the modern standard is **RAM live first, then dead-box disk**.)

4.3 Imaging vs cloning; formats; write blockers; validation (recap)

- **Imaging** = create a *file* (raw/dd, **E01/EWF**, **AFF**) representing the device; **cloning** = disk-to-disk copy. Prefer imaging for evidence. (Formats: FOR_03 §5.3.)
- **Write blockers** (hardware = gold standard) prevent any write to the evidence. (FOR_03 §5.2.)
- **Data validation** = prove the acquired copy equals the source and stays unchanged, by **hashing** (MD5 128-bit / SHA-1 160-bit / SHA-256 256-bit; compute at acquisition, re-verify later; use two algorithms). (FOR_03 §5.4, incl. the collision/SSD-GC caveats.)

§5. Hash Algorithms, File Extensions & Signatures, Registry, Recovery, Disks

These are the "toolbox" facts Unit 8 lists; each is treated fully elsewhere — here is the exam-ready compression with pointers.

5.1 Hash values (see FOR_03 §5.4)

MD5 (128-bit), SHA-1 (160-bit), SHA-256 (256-bit). Hash = the digital **seal/fingerprint**; avalanche effect; MD5/SHA-1 collision-broken (still fine against accidental change; use SHA-256 or dual hashes for defensibility). Uses: **integrity verification**, **de-duplication**, and **hash-set filtering** (NSRL known-good hashes to *exclude* OS files; known-bad hash sets to *flag* contraband like CSAM without opening files).

5.2 File extensions vs file signatures (see FOR_03 §6.3 for the full table)

Extension = renamable name; signature (magic bytes) = true type. Must-know headers: **JPEG** **FF D8 FF**, **PDF** **%PDF(25 50 44 46)**, **PNG** **89 50 4E 47**, **ZIP/Office** **PK(50 4B 03 04)**, **GIF** **GIF87a/GIF89a**, **EXE** **MZ(4D 5A)**, **ELF** **7F 45 4C 46**. Signature analysis detects extension-mismatch masquerading.

5.3 Windows Registry as evidence (see FOR_02 §5)

The registry is an **unintentional activity log**. High-value keys: **USBSTOR / MountedDevices / MountPoints2** (device history + user attribution), **UserAssist** (GUI program execution, ROT-13), **ShellBags** (folders browsed), **RecentDocs / RunMRU / TypedPaths / WordWheelQuery** (files, run commands, search terms), **Run/RunOnce/Services** (persistence), **TimeZoneInformation** (interpret all timestamps). Keys carry a **LastWrite** time; values do not. Cite this in any "registry as evidence" answer.

5.4 Data recovery techniques (see FOR_03 §6)

Undelete (metadata survives) → **carving** (content-based, no metadata) → **journal/shadow-copy** recovery (`$LogFile`, `$UsnJrnl`, **Volume Shadow Copies**). Deleted-file mechanics differ by FS (FAT `0xE5`; NTFS MFT unallocated; ext3/4 zero pointers). **SSD TRIM often destroys deleted data autonomously** (FOR_02 §2.2).

5.5 Windows disks, files and partitions — MBR vs GPT (a guaranteed compare)

Aspect	MBR (Master Boot Record)	GPT (GUID Partition Table)
Era / firmware	Legacy BIOS	UEFI
Location	First sector, LBA 0 (512 bytes): 446 B bootstrap + 64 B partition table (4×16) + 55 AA signature	GPT header at LBA 1 ; partition entries follow; a protective MBR at LBA 0 for backward compat
Max partitions	4 primary (or 3 primary + 1 extended → logical drives)	128 partitions typically (no extended/logical hack)
Max disk size	~2 TB (32-bit LBA × 512 B)	~9.4 ZB (64-bit LBA)
Redundancy	None (single MBR — a corrupt MBR loses the map)	Backup GPT header + table at end of disk ; CRC32 on header/table → integrity + recovery
Partition IDs	1-byte type code	GUID type + unique partition GUID
Forensic points	MBR is a single easily-imaged sector; MBR bootkits ; recover the 4 entries; disk signature at offset 440	Examine backup GPT to detect tampering; recover deleted partitions from the backup table; partition GUIDs aid identification

MCQ nuggets: MBR = 512 bytes, **55 AA** boot signature, max 4 primary partitions, 2 TB limit; GPT = UEFI, 128 partitions, huge size, redundant + CRC-checked. The **partition table lives in the MBR (LBA 0)**; GPT keeps a backup at the end of the disk.

Also relevant: **partitions vs volumes**, **slack space** (file/RAM/volume slack — FOR_02 §5.9), unallocated space, and **inter-partition/boot-record slack** as hiding places.

5.6 Email tracing, social-media analysis, steganalysis (see FOR_03 §6.5, §9, §1.5)

- **Email tracing:** read `Received:` headers **bottom→top** to the originating IP; check **SPF/DKIM/DMARC**, `From` vs `Return-Path/Message-ID/Reply-To`; then ISP legal process + §65B/§63 preservation. (Full walkthrough: FOR_03 §9.)
- **Social-media analysis:** documented open-source capture (URL + timestamp + hash) → platform legal process → device app-database forensics. (FOR_03 §1.5.)
- **Steganalysis:** statistical (chi-square, RS), signature, structure/size, comparison-to-original. (FOR_03 §6.5.)

§6. Forensic Report Writing and Expert Testimony

This closes the process (Phase 6). It is easy marks because it is a **list of qualities and contents** — and a CS graduate typically neglects it. Learn the report contents and the courtroom rules.

6.1 The forensic report — qualities and contents

Qualities (state these): accurate, objective/impartial, complete, clear (plain language), reproducible, defensible, and within the examiner's competence. Report facts and findings; keep opinion clearly separated and supported.

Standard contents (as a list — reproduce in the exam):

1. **Case/administrative details** — case/FIR number, requesting agency, examiner name & qualifications, dates.
2. **Authorisation & scope** — the warrant/authority and what was asked.
3. **Items received / chain of custody** — exhibit list, description, serial numbers, seal condition on receipt, **hash values**.
4. **Tools and methods** — hardware/software used **with versions**; write blocker used; SOP followed.
5. **Acquisition details** — type of image, source & image **hashes**, verification result.
6. **Findings** — the objective facts recovered (files, artifacts, timeline), with locations.
7. **Analysis / interpretation** — what the findings mean, correlations, timeline reconstruction.
8. **Conclusions / opinion** — clearly labelled as opinion, tied to findings, with limitations.
9. **Exhibits/appendices** — screenshots, extracted files, hash lists, logs.
10. **Declaration & signature** — including the **§65B IEA / §63 BSA certificate** for the electronic record.

6.2 Expert testimony (link to FOR_01 §3.3 duties of an expert)

- Expert opinion is admissible under **§45 Indian Evidence Act 1872 (now §39 BSA 2023 [△](#) verify)**; it is **advisory — the court is not bound** and must apply its own mind.
- Reports of certain **Government Scientific Experts** may be used **without calling the expert** (old **CrPC §293**, now the corresponding **BNSS** provision [△](#) verify number) — the GEQD, Chemical Examiner, FSL Director, etc. (FOR_01 §2.5).
- **The expert's primary duty is to the court**, not to the party that engaged them; testify **only within your field**; explain in plain language; disclose assumptions, limitations and anything detracting from the opinion; be ready to produce **case notes, tool logs, hashes and SOPs**.
- **Cross-examination targets:** the **chain of custody**, whether a **write blocker** was used, **hash verification**, tool validation, and examiner competence — which is exactly why the report must pre-answer all of them. **Never overstate** ("consistent with" vs "conclusively proves").

§6.3 Likely exam questions (Part A)

Marks	Question
5	List the characteristics of digital evidence.
5	State the ACPO principles of digital evidence.
5	Distinguish MBR and GPT.
5	Distinguish physical, logical and sparse acquisition.
5	What are the contents of a forensic report?
10	Explain the phases of the digital forensic process model with a diagram.
10	Discuss data acquisition methods (physical/logical/sparse, live/static) and when each is used.
10	Discuss forensic report writing and the duties of an expert witness in an Indian court.
15	Describe the complete lifecycle of digital evidence from identification to presentation, referencing ACPO principles, hashing, chain of custody and the §65B/§63 certificate.

MCQ traps (Part A)

- **ACPO has 4 principles**; Principle 1 = "no action should change the data".
- Process model (6): **Identification** → **Preservation** → **Collection** → **Examination** → **Analysis** → **Presentation**; **NIST condenses to 4** (Collection, Examination, Analysis, Reporting).
- **MBR max 4 primary partitions**, ~2 TB, **55 AA**; **GPT 128 partitions**, **UEFI**, **backup table + CRC**.
- **Logical acquisition does not capture deleted data**; **physical (bit-stream) does**.
- **Expert opinion is advisory** and admissible under IEA §45 / BSA §39 (△ verify §39).
- Admissibility needs the **§65B IEA / §63 BSA certificate**.

PART B — MOBILE COMPUTING (TP-II Unit 7, engineering half)

This half overlaps CORE networks; keep it tight. The examiner wants the **cellular concepts** (cells, reuse, handoff), the **generation timeline (1G→5G)**, **GSM vs CDMA**, and **mobile databases / M-business**. Then Part C turns to the forensic payoff.

§7. Mobile Connectivity and Cellular Concepts

7.1 Cells and the cellular idea

Concept: radio spectrum is scarce. Instead of one giant transmitter, the area is divided into small **cells**, each served by a **base station (BTS/Node B/eNodeB/gNodeB)** at low power, so the **same frequencies can be reused** in non-adjacent cells. This **frequency reuse** multiplies capacity — the founding idea of cellular telephony.

Term	Meaning
Cell	Geographic area served by one base station; drawn as a hexagon (tessellates without gaps/overlap)
Frequency reuse	Reusing the same channel set in cells far enough apart to avoid co-channel interference; reuse factor / cluster size (e.g. 3, 7)
Cell splitting / sectoring	Split a congested cell into smaller cells, or sector the antenna (e.g. 3×120°) to add capacity
Handoff / handover	Transferring an active call from one cell to the next as the user moves (hard handoff = break-before-make, GSM; soft handoff = make-before-break, CDMA)
Roaming	Using the network outside the home network's coverage
MSC / HLR / VLR	Mobile Switching Centre; Home & Visitor Location Registers — track subscriber location and route calls

Wireless delivery technologies & switching methods: early cellular used **circuit switching** (a dedicated channel for the call, like a landline). Data services moved to **packet switching** (**GPRS/EDGE** overlay on GSM), and modern networks (**LTE/5G**) are **all-IP packet-switched**, carrying even voice as packets (**VoLTE**). Switching evolution — *circuit* → *packet* → *all-IP* — is worth one sentence.

7.2 Cellular generations (1G → 5G) — the timeline

Gen	Era	Switching	Key technology	Data
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1G	1980s	Circuit	Analog (AMPS)	Voice only
2G	1990s	Circuit (+packet overlay)	Digital — GSM (TDMA/FDMA) and CDMA (IS-95) ; SMS	~kbps; GPRS (2.5G) ~ up to ~114 kbps; EDGE (2.75G) ~up to ~384 kbps
3G	2000s	Packet + circuit	UMTS/WCDMA , CDMA2000 ; HSPA/HSPA+	~Mbps mobile broadband
4G	2010s	All-IP packet	LTE / LTE-Advanced (OFDMA); VoLTE	tens–hundreds of Mbps
5G	2019+	All-IP	NR (New Radio), mmWave, massive MIMO, network slicing	Gbps, ultra-low latency, massive IoT

7.3 GSM vs CDMA (a classic compare)

Aspect	GSM	CDMA
Access method	TDMA + FDMA (time + frequency slots)	Code Division — all share the band, separated by unique codes
Subscriber identity	SIM card — identity is portable between handsets	Historically handset-bound (no removable SIM in classic CDMA)
Handoff	Hard (break-before-make)	Soft (make-before-make)
Global reach	Dominant worldwide	Fewer markets
Forensic note	SIM is a discrete evidence item (see §9); IMSI/ICCID on the SIM	Subscriber data tied to the device; no SIM to seize (in classic CDMA)

7.4 Mobile information access devices, internetworking standards, WAP

- **Devices:** smartphones, tablets, feature phones, wearables, IoT/M2M modules, in-vehicle units.
- **Mobile data internetworking standards:** **GPRS/EDGE (2G data)**, **UMTS/HSPA (3G)**, **LTE/5G-NR**; local wireless **Wi-Fi (802.11)**, **Bluetooth**, **NFC**, **Zigbee**.
- **WAP (Wireless Application Protocol)** — the early stack for delivering web-like content to constrained phones: layered as **WAE / WSP / WTP / WTLS / WDP** over the bearer, with **WML** markup and a **WAP gateway** translating between WAP and HTTP. Largely historical (superseded by full mobile browsers) but examinable as a named stack. ⚠ verify layer names before writing all five.

7.5 Mobile databases, tools/technology, and M-business

Mobile database — concept and needs: a database that runs on or synchronises with mobile devices under **intermittent connectivity**, **limited power/storage**, and **mobility**. Requirements: **local storage + replication/synchronisation** with a central server, **conflict resolution**, **caching**, **disconnected operation**, and **security/encryption** on a device that is easily lost.

Concept	Note
Replication & synchronisation	Local copy syncs with server when connected; conflict resolution (last-writer-wins, merge) needed

Caching / hoarding	Pre-fetch likely-needed data for offline use
Transaction models	Long-lived/disconnected transactions; optimistic concurrency suits mobility
On-device engines	SQLite (Android/iOS default), Realm; server side syncs to the enterprise DB
Forensic payoff	SQLite databases are where the evidence is on phones — SMS, call logs, contacts, chat apps, browser history all live in SQLite files (see §11)

M-business (mobile commerce): conducting commerce via mobile devices — mobile banking/payments (**UPI**, wallets), m-ticketing, location-based services, m-marketing. Enablers: secure mobile payment, location awareness, ubiquity, personalisation. **Forensic relevance:** payment-app databases, UPI transaction records, location history — heavily used in fraud cases (FOR_03 §1.6).

§7.6 Likely exam questions (Part B)

Marks	Question
5	What is frequency reuse? Why are cells drawn as hexagons?
5	Distinguish hard and soft handoff.
5	Distinguish GSM and CDMA.
5	What is a mobile database and what challenges does mobility create?
10	Trace the evolution of cellular networks from 1G to 5G.
10	Explain mobile database synchronisation and its forensic significance.
15	Describe the architecture of a cellular mobile network (cells, base stations, MSC/HLR/VLR, handoff) and explain how mobility and switching evolved from 1G to 5G.

MCQ traps (Part B)

- **GSM uses TDMA/FDMA + SIM; CDMA uses code division, historically SIM-less.**
- **GPRS/EDGE are 2.5G/2.75G packet data; UMTS/WCDMA = 3G; LTE = 4G; NR = 5G.**
- **1G analog, 2G digital (+SMS).**
- **Cells are hexagonal** (tessellation, uniform coverage).
- **Soft handoff = CDMA (make-before-break); hard handoff = GSM.**
- **SQLite** is the default on-device mobile database.

PART C — MOBILE FORENSICS (TP-II Unit 7, forensic half)

§8. Mobile Forensics — Concept and Challenges

Concept: the science of recovering digital evidence from mobile devices under forensically sound conditions. A phone is a **richer, harder** exhibit than a PC: it holds calls, SMS, chats, photos (with GPS), app data, location history and health data — but it is **always-on, always-networked, encrypted by default, and endlessly varied** in OS/model.

Why it is harder than disk forensics:

- **Always connected** → risk of **remote wipe** / **new data overwriting** → **isolate immediately** (Faraday bag / airplane mode, keep charged).

- **Encryption by default** (modern Android/iOS full-disk/file-based encryption) → data unreadable without the passcode; **locked vs unlocked (AFU/BFU) state matters** hugely.
- **Proprietary, fast-changing** hardware/OS/connectors → tool support lags.
- **Volatile, hard to write-block** — you cannot easily attach a hardware write blocker to a soldered eMMC/UFS; acquisition inevitably interacts with the live device (document per ACPO P2).
- **State: BFU vs AFU — Before First Unlock** (keys not yet derived, very little accessible) vs **After First Unlock** (keys in memory, far more data recoverable). Keep an AFU phone **powered and unlocked-alive** if lawful.

§9. SIM, Handset and Card Evidence; IMEI vs IMSI

9.1 The three evidence sources in a phone

Source	What it holds
Handset (internal memory)	The bulk: OS, apps, SQLite databases (SMS, call log, contacts, chat, browser), photos/videos with EXIF/GPS, location history, account tokens
SIM/USIM card	IMSI, ICCID , operator data, phonebook (ADN) , SMS (some stored on SIM) , last dialled numbers (LND), location info (LAI/TMSI)
Removable media (SD card)	Photos, videos, downloads, app data, sometimes backups — image separately with a write blocker

9.2 IMEI vs IMSI vs ICCID — memorise the distinction (guaranteed MCQ)

Identifier	Identifies	Length / format	Where it lives
IMEI (International Mobile Equipment Identity)	The handset/device	15 digits (TAC + serial + check digit); dial *#06# to display	In the phone hardware
IMSI (International Mobile Subscriber Identity)	The subscriber (the SIM/account)	up to 15 digits = MCC (3) + MNC (2–3) + MSIN	On the SIM/USIM
ICCID (Integrated Circuit Card ID)	The SIM card itself (serial number)	up to 19–20 digits	Printed on and stored in the SIM
MSISDN	The phone number		Network (HLR), not necessarily on the SIM

The one-liner: IMEI = the device; IMSI = the subscriber; ICCID = the SIM card; MSISDN = the number. Swap a stolen phone's SIM and the **IMEI stays** (traces the handset) while the **IMSI changes** — which is exactly how stolen-handset tracing works. **CEIR** (Central Equipment Identity Register) / the sancharsaathi.gov.in portal blocks stolen IMEIs in India △ verify name before quoting.

9.3 CDR analysis — Call Detail Records (a high-value topic)

Concept: a **CDR** is the **metadata record a telecom operator keeps for every call/SMS/data session** — *not* the content, but who/when/where/how long. Subpoena the operator; CDRs are a mainstay of Indian investigations (they place a suspect near a scene and map their network).

Typical CDR fields:

Field	Use
Calling & called numbers (A-party / B-party)	Who contacted whom
Date, time, duration	Timeline; pattern of life
Call type (voice/SMS/data), direction (MO/MT)	Nature of contact
IMEI	Which handset was used (catches SIM-swapping across handsets)
IMSI	Which SIM/subscriber
Cell ID / LAC + tower location, azimuth/sector	Approximate location of the phone at the time — the crux of CDR-based placing of a suspect
First/last cell	Movement during a call

What CDR analysis achieves: place a suspect near the crime scene at the time (tower/cell location), establish contact between conspirators, build a **timeline and social-network graph**, detect **SIM-swap** (same IMEI, different IMSI) or **handset-swap** (same IMSI, different IMEI), and corroborate other evidence. **Limitations to state:** cell location is **approximate** (coverage area, not GPS), can be affected by tower load/handoff, and **CDR is metadata — content needs lawful interception** (IT Act §69) or on-device recovery.

9.4 Tower dump

A **tower dump** = all phones that connected to a given tower/cell in a time window — used to find an unknown suspect present at a scene. Privacy-sensitive; needs lawful authority.

§10. Mobile Acquisition Levels — the pyramid (learn the order)

A **classic question**: "Explain the levels of mobile data acquisition." Learn them from **least to most invasive** / **least to most complete** and note the trade-off (more data $\neq$ more risk/skill).

MOST data, MOST invasive/technical

- ▲
- 5. Chip-off — desolder the memory chip, read directly
 - 4. JTAG/ISP — connect to test/eMMC pads, read raw NAND
 - 3. Physical — bit-for-bit image of flash (deleted data)
 - 2. File system — copy the live file-system (app DBs, more than logical)
 - 1. Logical — API/backup extraction (contacts, SMS, call log, media)
 - 0. Manual — scroll & photograph the screen by hand
- ▼

LEAST data, LEAST invasive

Level	Method	Recovers	Notes / risk
Manual	Operate the phone, photograph screens	Only what is visible	Changes state; no deleted data; last resort or quick triage

Logical	Use the OS/backup API (e.g. ADB backup, iTunes-style backup, vendor sync)	Contacts, SMS, call logs, media, some app data	Fast, safe, tool-supported; usually no deleted data ; needs the device unlocked
File system	Pull the device's file-system (databases, caches, config)	App SQLite DBs, logs, more artifacts including some "deleted" rows in SQLite (unvacuumed/WAL)	Often needs root/exploit/agent; richer than logical
Physical	Bit-for-bit image of the flash	Everything incl. unallocated/deleted (if not encrypted)	The goal; hard on modern encrypted phones
JTAG / ISP	Connect to JTAG test points or In-System Programming (eMMC pads) to read raw memory	Full physical image without desoldering	Advanced; needed for damaged/locked devices
Chip-off	Desolder the NAND/eMMC/UFS chip and read it in a reader	Raw physical memory	Destructive , expensive, expert-only; encryption still blocks reading without keys

Key modern caveat: on encrypted phones, **physical/JTAG/chip-off give you ciphertext** unless you also have the key (from an AFU state, a known passcode, or an exploit). So **logical/file-system acquisition from an unlocked/AFU device is often more valuable than a chip-off of a locked one**. Say this — it is the sophisticated point examiners reward.

SQLite recovery angle: even "deleted" chat/SMS rows often survive in **SQLite free-pages, WAL/journal files, and unvacuumed space** — a major reason file-system acquisition matters. Deleted rows are recoverable by carving the SQLite structures.

§11. Android and iOS Artifacts

Artifact	Android	iOS
Storage/FS	Historically ext4/F2FS; user data in <code>/data/data/<app>/</code>	APFS ; data under <code>/private/var/mobile/</code>
Encryption	FBE (file-based encryption), earlier FDE	Data Protection (per-file keys tied to passcode + Secure Enclave)
Contacts	<code>contacts2.db</code> (SQLite)	<code>AddressBook.sqlitedb</code>
SMS/MMS	<code>mmssms.db</code>	<code>sms.db</code>
Call log	<code>calllog.db</code> / <code>contacts2.db</code>	<code>call_history.db</code> / <code>CallHistory.storedata</code>
Chat apps	Per-app SQLite in <code>/data/data/<pkg>/databases/</code> (e.g. WhatsApp <code>msgstore.db</code> , often encrypted)	Per-app in the app container; WhatsApp <code>ChatStorage.sqlite</code>
Browser	Chrome SQLite (<code>History</code>)	Safari <code>History.db</code>
Photos/media	DCIM + <code>MediaStore</code> ; EXIF incl. GPS	Photos library + <code>Photos.sqlite</code> ; EXIF/GPS
Location	Google location history, cell/Wi-Fi caches	<code>cache.sqlite</code> , frequent locations, <code>KnowledgeC.db</code> (device usage timeline)

Accounts/ tokens	<code>accounts.db</code> , app credentials	Keychain (Secure Enclave-protected)
Backups	ADB backup, cloud (Google)	iTunes/Finder backup (may be encrypted), iCloud
System usage	<code>usagestats</code> , logcat	KnowledgeC.db , <code>powerlog</code> , unified logs

The recurring theme: almost every phone artifact is a SQLite database. An Android/iOS answer that says "SMS in `mmssms.db/sms.db`, calls in `callog.db/call_history.db`, contacts, and app chat DBs are SQLite files recovered by file-system acquisition, with deleted rows carved from SQLite free-pages/WAL" scores well. ⚠ verify exact filenames per OS version before quoting them as absolute — they drift across versions; the **concept** (SQLite app DBs + EXIF/GPS + location DBs like KnowledgeC) is the examinable part.

§11.1 Likely exam questions (Part C)

Marks	Question
5	Distinguish IMEI, IMSI and ICCID.
5	What is a CDR? List its main fields.
5	What evidence can be recovered from a SIM card?
5	Why must a seized phone be placed in a Faraday bag?
5	Distinguish logical and physical acquisition of a mobile phone.
10	Explain the levels of mobile forensic acquisition from manual to chip-off.
10	Explain CDR analysis and how it places a suspect at a scene, with its limitations.
10	Describe the artifacts recoverable from an Android or iOS device.
15	Discuss the challenges of mobile forensics (encryption, connectivity, BFU/AFU state) and explain, with the acquisition-level pyramid, how you would forensically acquire a locked smartphone.

MCQ traps (Part C)

- **IMEI = equipment/handset (15 digits, *#06#); IMSI = subscriber (on SIM); ICCID = SIM serial.**
- **Same IMEI + different IMSI = SIM swap; same IMSI + different IMEI = handset swap.**
- **CDR is metadata (who/when/where), not call content.** Content needs §69 interception.
- **Chip-off is destructive; JTAG/ISP is non-destructive but advanced; both yield ciphertext if encrypted.**
- **Faraday bag** prevents remote wipe / network changes.
- **Cell/tower location is approximate**, not GPS-precise.
- Most phone artifacts are **SQLite**; deleted rows survive in **WAL/free-pages**.

§12. Forensic Tools — the reference table (know 2–3 lines on each)

A "name and describe forensic tools" question is common, and every walkthrough answer is stronger if you name the right tool for the step. Learn the **category** → **tool** → **one-line use**.

Tool	Type / vendor	Primary use
EnCase	Commercial suite (OpenText/Guidance)	Full disk imaging (E01), analysis, keyword search, reporting; long courtroom pedigree; EnScript automation
FTK (Forensic Toolkit) + FTK Imager	Commercial (Exterro/AccessData)	Indexed searching, analysis; FTK Imager is the free, ubiquitous imaging/preview tool (raw/E01/AD1, memory capture)
Autopsy / The Sleuth Kit (TSK)	Open source (Brian Carrier)	GUI (Autopsy) over TSK CLI; file-system analysis, carving, timeline, keyword, hash sets — the standard free suite
X-Ways Forensics	Commercial (lightweight)	Fast, low-footprint disk analysis and imaging
Cellebrite UFED	Commercial (mobile)	Industry-standard mobile acquisition (logical/file-system/physical), lock bypass, app parsing
Magnet AXIOM / Oxygen Forensic Detective / MSAB XRY	Commercial	Mobile + computer + cloud artifact analysis; strong app/cloud parsing
Wireshark / tcpdump	Open source (network)	Packet capture/analysis (PCAP); protocol dissection; the network sniffer/analyser
Volatility / Rekall	Open source (memory)	Memory (RAM) forensics — processes, network, injected/fileless malware, keys
dd / dcfldd / dc3dd	Open source (imaging)	Command-line bit-stream imaging ; dcfldd/dc3dd add hashing & progress
HashCalc / md5sum / sha256sum / hashdeep	Utilities	Compute/verify hash values for integrity
RegRipper / Registry Explorer	Open source/free	Parse Windows registry hives into readable artifacts
PhotoRec / Scalpel / Foremost	Open source	File carving from unallocated space
Plaso / log2timeline	Open source	Super-timeline creation across many artifact types
Bulk Extractor	Open source	Scan images for emails, cards, URLs without parsing the FS
NSRL (RDS)	NIST hash set	Known-good hashes to filter out OS/software files
SANS SIFT / Kali / CAINE	Forensic Linux distros	Bootable toolkits bundling the above

Quick mapping to memorise: imaging → **FTK Imager / dd / EnCase**; **disk analysis** → **Autopsy/TSK, EnCase, FTK, X-Ways**; **memory** → **Volatility**; **network** → **Wireshark**; **mobile** → **Cellebrite UFED / XRY / AXIOM**; **registry** → **RegRipper**; **carving** → **PhotoRec/Scalpel**; **hashing** → **hashdeep/sha256sum**; **timeline** → **Plaso**.

§12.1 Likely exam questions

Marks	Question
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